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Making fuzzy description logic more general☆

Petr Hájek*

Institute of Computer Science, Academy of Sciences of the Czech Republic, 182 07 Prague, Czech Republic

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Abstract

A version of fuzzy description logic based on the basic (continuous t-norm based) fuzzy predicate logic BL is presented. Problems of satisfiability, validity and subsumption of concepts are discussed and reduced to problems of fuzzy *propositional* logic known to be decidable for any continuous t-norm. For Łukasiewicz t-norm some stronger results are obtained.
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1. Introduction

Description logic has been extensively studied in recent years; the handbook [2] is an up-to-date reference. (See also [1].) It is known to be a fragment of classical predicate calculus and has several variants; we restrict ourselves to that named ALC. Its language consists of unary predicates $A_1, \ldots$ (atomic concepts), binary predicates $R_1, \ldots$ (roles), variables $x_1, \ldots$ and constants $a_1, \ldots$. Concepts are built from atomic concepts using connectives $\wedge$, $\vee$, $\neg$ and quantification constructs denoted $\forall R.C$, $\exists R.C$. Think of instances of atomic concepts as formulas $A_i(t)$ (t being a fixed constant or variable); this extends to (instances of) concepts defined by connectives. Furthermore, when translated to the

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* Tel.: +42 2 6884 244; fax: +42 2 8585 789.
E-mail address: hajek@cs.cas.cz.

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classical predicate calculus,

$(\forall R.C)(t)$ is to be read as $(\forall y)(R(t, y) \to C(y))$

$(\exists R.C)(t)$ as $(\exists y)(R(t, y) \wedge C(y))$.

(Cf. [2, p. 38, 72] or [4]. For each construct always a new variable y is used.)

Using $\wedge$, $\vee$, $\neg$ is a possible choice of basic connectives. Needless to say, all connectives definable in classical logic from the basic ones are also available, in particular the implication; classically, $C \to D$ is of course equivalent to $\neg C \vee D$, i.e., $C(x) \to D(x)$ is classically equivalent to $\neg C(x) \vee D(x)$.

Axioms may have the form $C(a_i)$ (C a concept), $R(a_i, a_j)$ (R a role) and inclusion of concepts denoted $C \sqsubseteq D$, which translates as $(\forall x)(C(x) \to D(x))$ (See [2, p. 51].).[1] Similarly for inclusion of roles. We simplify a little and will disregard (a priori) axioms. Typical *problems* are:

satisfiability (alternative name: coherence): does $C(a)$ have a model?
validity: is $(\forall x)C(x)$ true in all models? In particular,
subsumption: is $(\forall x)(C(x) \to D(x))$ true in all models?[2]

Both problems are *decidable* (and are PSPACE-complete; furthermore, satisfiability (coherence) is decidable in linear space and exponential time—see [14]).

It is very natural, due to the intended applications, to make this system (and related systems) fuzzy.[3] The handbook [2] refers to papers [15–17]. These are rather non-trivial papers but they restrict themselves to a rather poor apparatus of fuzzy logic: just min and max as $\wedge$ and $\vee$, $1 - x$ (Łukasiewicz negation) and, when defining the semantics of $\forall R.C$, they use $\max(1 - x, y)$ (Kleene–Dienes implication).[4] The use of "minimalist" fuzzy apparatus has some technical advantages but makes the system(s) unnecessarily weak.

Technically most developed are t-norm based fuzzy logics; the corresponding propositional and predicate calculi are elaborated in the monograph [6] (basic fuzzy logic BL and several stronger logics, notably the Łukasiewicz, Gödel and product logics).

The reader not familiar with these logics, their standard and general semantics, may consult Appendix A for a quick survey.

Note that in this paper we deal only with the *standard* semantics, given always by a fixed continuous t-norm; our set of truth values is just the unit real interval [0, 1].

Our aim is to reformulate the description logic (ALC-like) using this apparatus, formulate some typical problems and prove some preliminary results; notably, we prove that both satisfiability of a concept (in a finite model) and validity of a concept (in all finite models) is decidable. We also show that "finite" may be replaced by "witnessed"; witnessed models (both finite and infinite) present a new kind of models, apparently rather interesting ones. Some first results on them are presented. We also correct an (inessential) error in [17].

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[1] One has to distinguish carefully the *concept* $C \to D$ from the *axiom* $C \sqsubseteq D$, the latter saying that all objects having C have D, i.e., all objects satisfy the implication (concept) $C \to D$.

[2] Our definition is trivially equivalent to that in [2, p. 62] saying that in each interpretation $\mathcal{I}$ the set $C^{\mathcal{I}}$ of objects satisfying C in $\mathcal{I}$ is a subset of $D^{\mathcal{I}}$.

[3] Trivial example: the concept of a person, all of whose children are tall; and of a person having at least one tall child.

[4] Ref. [2] refers also to an older paper [18] dealing with different logical systems; [10] considers use of various conjunctions and implications. Ref. [11] offers fuzzy interpretation of concepts using general fuzzy semantics of connectives but presents no results about it.

We define the semantics of concepts in full analogy to the classical case but understand the defining formulas of the predicate calculus using their meaning in fuzzy logic. Our main technical contribution is the systematic reduction of problems of satisfiability and validity of concepts to problems of satisfiability and validity of some theories in *propositional* logic, problems that are known to be decidable. Our method of proving decidability is *uniform*, i.e., does not depend on the choice of a continuous t-norm as defining the particular fuzzy logic—in contradistinction to the proof in [17], which can be generalized for the Łukasiewicz logic but not for any other continuous t-norm logic. The paper should serve as a programme of further deeper research, using more deeply results from the existing literature on description logic in combination with results of mathematical fuzzy logic.

2. Main notions; an algorithm

This section contains basic definitions and prepares the promised reduction of validity/satisfiability of concepts to some problems of fuzzy propositional calculus. When doing this the notion of a witnessed model naturally emerges.

Recall that basic connectives of our fuzzy logic are conjunction and implication (& and $\rightarrow$). This leads us to the following definition:

Definition 1. (1) Given atomic concepts $A_1, \ldots, A_n$ and roles $R_1, \ldots, R_m$, *concepts* are defined as follows: atomic concepts are concepts; $\bar{0}$ (falsum) is a concept; if C, D are concepts then so are $C\&D$, $C \rightarrow D$; if C is a concept and R a role then $\forall R.C$ and $\exists R.C$ are concepts.

(2) For each variable or constant t, the *instance* $C(t)$ of a concept is defined as follows: $A(t)$ is the atomic formula in which A is understood as unary predicate; $\bar{0}(t)$ is $\bar{0}$; $(C \rightarrow D)(t)$ is $C(t) \rightarrow D(t)$, similarly $(C\&D)(t)$. Furthermore, $(\forall R.C)(t)$ is $(\forall y)(R(t, y) \rightarrow C(y))$ and $(\exists R.C)(t)$ is $(\exists y)(R(r, y)\&C(y))$ where y is a variable not occurring in $C(t)$.

(3) *Nesting of quantifiers* in C (or $C(t)$) is defined inductively: $nest(C) = 0$ for atomic C, $nest(C\&D) = \max(nest(C), nest(D))$, $nest(\forall R.C) = nest(C) + 1$, similarly for $\rightarrow, \exists$.

(4) *Generalized atoms* are instances of quantified concepts, i.e., $(\forall R.C)(t)$, $(\exists R.C)(t)$. Evidently, each instance $C(t)$ of any concept is a propositional combination of some atoms and generalized atoms; the latter will be called *generalized atoms of* $C(t)$.

Remark 1. Recall the definable connectives $\neg, \wedge, \vee$ (see Appendix A); thus if C, D are concepts then also $\neg C$, $C \wedge D$, $C \vee D$ are concepts.

Example 1. $\forall R_1.\exists R_2.C\&\forall R_2.D$ is a concept; its instance with a constant a is $(\forall R_1.\exists R_2.C)(a)\&(\forall R_2.D)(a)$, i.e.,

$$[(\forall y)(R_1(a, y) \rightarrow (\exists z)(R_2(y, z)\&C(z)))]\&(\forall y)(R_2(a, y) \rightarrow D(y)).$$

Generalized atoms of our instance of this concept are $(\forall R_1.\exists R_2.C)(a)$ and $(\forall R_2.D)(a)$.

Definition 2. Assign to each generalized atom $G(a)$ (G atomic concept or a quantified concept, a a constant) a propositional variable $p_{G,a}$. Extend this to all instances $C(a)$ of concepts by defining

$prop(G(a)) = p_{G,a}$ (for any generalized atom), $prop(\bar{0}(a)) = \bar{0}$,
$prop((C\&D)(a)) = prop(C(a))\&prop(D(a))$ and similarly
$prop((C \to D)(a)) = prop(C(a)) \to prop(D(a))$.

If T is a set of formulas (instances of concepts) let $prop(T)$ be the set of all $prop(\alpha)$, $\alpha \in T$.

Thus, continuing Example 1, $prop(\forall R_1.\exists R_2.C\&\forall R_2.D)$ is just the conjunction of two propositional variables, namely $p_{\forall R_1.\exists R_2.C}\&p_{\forall R_2.D}$.

Now we describe an algorithm (in the style of cited papers on description logic) assigning to each (closed) instance $C_0(a_0)$ of a concept C_0 with a constant a_0 a finite "witnessing" theory $T(C_0(a_0))$ important in what follows. The construction takes steps $0, \ldots, n$ where n is the degree of nesting of quantifiers in C_0. Each step will *process* some generalized atoms, *produce* some new constants and new axioms and *transfer* some instances of concepts for processing in the next step. The instances produced in step i will have degree of nesting $n - i$; after step n is completed the algorithm will stop. The theory T in question consists of all axioms produced in all steps. In details:

Definition 3. Given $C_0(a_0)$, step 0 just transfers it to further processing in step 1; the constant a_0 has level 0. For $i > 0$ step i processes generalized atoms of formulas transferred from step $i - 1$; they have the form $(QR.C)(b)$ where Q is $\forall$ or $\exists$, R is a role, C a concept with nesting degree $\leqslant n - i$ and b is a constant of level i. For each generalized atom α in question do the following:

If α is $(\forall R.C)(b)$ then produce a new constant d_α and the axiom

$$(\forall R.C)(b) \equiv (R(b, d_\alpha) \to C(d_\alpha)).$$

If α is $(\exists R.C)(b)$ then produce d_α and axiom

$$(\exists R.C)(b) \equiv (R(b, d_\alpha)\&C(d_\alpha)).$$

In both cases call the generated axiom the *witnessing axiom* for α and d_α a constant *belonging to* R, b.

After this is done for all α in question (in the present step) consider each α once more and do the following:

If α is $(\forall R.C)(b)$ and d_β is any constant belonging to R, b and different from d_α, produce the axiom

$$(\forall R.C)(b) \to (R(b, d_\beta) \to C(d_\beta)).$$

Similarly for α being $(\exists R.C)(b)$, produce

$$(R(b, d_\beta)\&C(d_\beta)) \to (\exists R.C)(b).$$

All produced axioms and constants are of level i. Formulas $C(d_\alpha)$, $C(d_\beta)$ from the axioms produced at this level and having positive nesting of quantifiers are transferred for processing in the next level (if any). Observe that finitely many constants, axioms and formulas for the next processing have been generated and that all the generated formulas for the next processing have nesting of quantifiers $\leqslant n - i$. This completes the definition.

Clearly, the algorithm stops after the n steps are performed. As stated above, T is the set of all axioms produced. We shall investigate properties of T and of its propositional counterpart $prop(T)$.

From now on, fix a continuous t-norm $*$ (and its residuum $\Rightarrow$); this gives the semantics of your fuzzy propositional and predicate calculus. In particular, each evaluation e of propositional atoms by truth values from the real interval $[0, 1]$ defines uniquely the truth value $e_*(\varphi)$ (given by e and $*$) of each propositional formula built from these atoms. We write $e(\varphi)$ if $*$ is clear from the context.

For each evaluation e of atoms of formulas in $prop(T)$ define an interpretation $\mathbf{M}_e$ of the predicate language as follows: M_e is the set of all constants occurring in formulas in T. For each atomic concept A let $r_A(d) = e(prop(A(d)))$ if $prop(A(d))$ occurs in $prop(T)$; otherwise $r_A(d) = 0$. Similarly, $r_R(c, d) = e(prop(R(c, d)))$ if $prop(R(c, d))$ occurs in $prop(T)$, otherwise $= 0$.

Lemma 1. *Let $*$ be a continuous* t-*norm. Assume that e is a $*$-model of $prop(T)$. Then for each formula $\alpha = (\forall R.C)(c)$ with $prop(\alpha)$ occurring in $prop(T)$,*

$$e(prop(\alpha)) = \|\alpha\|^*_{\mathbf{M}_e} = \|R(c, d_\alpha) \to C(d_\alpha)\|^*_{\mathbf{M}_e}$$

and for $\alpha = (\exists R.C)(c)$ with $prop(\alpha)$ occurring in $prop(T)$,

$$e(prop(\alpha)) = \|\alpha\|^*_{\mathbf{M}_e} = \|R(c, d_\alpha) \& C(d_\alpha)\|_{\mathbf{M}_e}.$$

Consequently, $\mathbf{M}_e$ is a finite $*$-model of T (in the sense of predicate logic) and $e_*(prop(C_0(a_0))) = \|C_0(a_0)\|^*_{\mathbf{M}_e}$.

Proof. The proof is by induction on the degree of the nesting of α. Assume α is $(\forall R.C)(b)$; the proof for $(\exists R.C)(b)$ is dual. First let nesting degree of α be 1, i.e., C does not contain any quantifiers. Then

$$\begin{aligned}\|(\forall R.C)(b)\|^*_{\mathbf{M}_e} &= \|(\forall y)(R(b, y) \to C(y)\|^*_{\mathbf{M}_e}) \\ &= \inf_{d \in \mathbf{M}_e} (\|R(b, d) \to C(d)\|^*_{\mathbf{M}_e}).\end{aligned}$$

Observe that $prop(R(b, d))$ occurs in $prop(T)$ if *and only if* d belongs to R, b in the sense of our algorithm; then $\|R(b, d)\|^*_{\mathbf{M}_e} = e(prop(R(b, d)))$ and $\|C(d)\|^*_{\mathbf{M}_e} = e(prop(C(d)))$ by the definition of $\mathbf{M}_e$ since $prop(R(b, d))$ is a propositional variable and $prop(C(d))$ is a propositional combination of propositional variables (due to the assumption on nesting).

In particular, for the constant d_α we get

$$e(prop((\forall R.C)(b))) = \|R(b, d_\alpha) \to C(d_\alpha)\|^*_{\mathbf{M}_e},$$

for any other constant d belonging to R, b we get

$$e(prop((\forall R.C)(b))) \leqslant \|R(b, d) \to C(d)\|^*_{\mathbf{M}_e}$$

(since e gives value 1 to all axioms concerning R, b) and if d does not belong to R, b then $e(prop(R(b, d))) = 0$, $\|R(b, d) \to C(d)\|^*_{\mathbf{M}_e} = 1$) and we get again

$$e(prop((\forall R.C)(b))) \leqslant \|R(b, d) \to C(d)\|^*_{\mathbf{M}_e}.$$

Thus $\inf_{d \in \mathbf{M}_e} \|R(b, d) \to C(d)\|^*_{\mathbf{M}_e} = e(prop((\forall R.C)(b)))$ and hence

$$e(prop((\forall R.C)(b))) = \|(\forall R.C)(b)\|^*_{\mathbf{M}_e}.$$

Similarly for nesting $(i+1)$, nesting i having been verified; now $C(d)$ is a propositional combination of atoms and generalized atoms with nesting $\leqslant i$ and for all such formulas the e-value equals to the $*$-value in $\mathbf{M}_e$. $\square$

Observe that the $\mathbf{M}_e$-value of $(\forall R.C)(c)$ is $*$-*witnessed* in the following sense:

Definition 4. For any $\mathbf{M}$, a closed formula $(\forall y)\varphi(y)$ is *-witnessed if $\|(\forall y)\varphi(y)\|^*_{\mathbf{M}}$ (which is the infimum of values of M-instances of $\varphi(y)$) is in fact equal to $\|\varphi(u)\|^*_{\mathbf{M}}$ for some $u \in M$, thus the infimum is in fact the *minimum*. Similarly for witnessed $\|(\exists y)\varphi(y)\|^*_{\mathbf{M}}$ (*maximum*). More generally, $(\forall y)\varphi(y, x_1, \ldots, x_n)$ is $*$-witnessed in $\mathbf{M}$ if for any choice $v_1, \ldots, v_n \in M$ of values of $x_1, \ldots, x_n$, the truth value $\|(\forall y)\varphi(y, v_1, \ldots, v_n)\|^*_{\mathbf{M}}$ equals $\|\varphi(u, v_1, \ldots, v_n)\|^*_{\mathbf{M}}$ for some $u \in M$. Similarly for $(\exists y)\varphi(y, x_1, \ldots, x_n)$ and maximum. $\mathbf{M}$ is *-*witnessed* if all quantified formulas are *-witnessed in $\mathbf{M}$.

Lemma 2. *Conversely, let* $\mathbf{M}$ *be a witnessed interpretation*; *expand the language by constants from* T *and expand* $\mathbf{M}$ *by interpreting the added constants as respective witnesses. For atoms* α *of formulas in* $prop(T)$ *define* $e_{\mathbf{M}}(\alpha) = \|\alpha\|^*_{\mathbf{M}}$. *Then* $e_{\mathbf{M}}$ *is a* $*$-*model of* $prop(T)$ *and hence* $e_{\mathbf{M}}(C_0(a_0)) = \|C_0(a_0)\|^*_{\mathbf{M}}$.

Proof. The constant a_0 is assumed to be interpreted in $\mathbf{M}$; in step $i+1$ assume the constants of level i have been interpreted in $\mathbf{M}$. For each generalized atom $\alpha = (\forall R.C)(c)$ processed in step i choose an element $u \in M$ witnessing that atom and interpret d_α by u (calling the expansion of $\mathbf{M}$ by these constants again $\mathbf{M}$). This means that $\|(\forall R.C)(c) \equiv (R(c, d_\alpha) \to C(d_\alpha))\|_{\mathbf{M}} = 1$ and analogously for $\alpha = (\exists R.C)(c)$. Also the remaining axioms produced in this step are obviously true in $\mathbf{M}$. Thus setting, for each atom and generalized atom α, $e(prop(\alpha)) = \|\alpha\|^*_M$, we get $e_*(prop(\varphi)) = \|\varphi\|^*_{\mathbf{M}}$ for each propositional combination of atoms and generalized atoms occurring in T. In particular, the (expanded) model $\mathbf{M}$ is a model of T and e is a model of $prop(T)$. $\square$

Remark 2. Let $\mathbf{M}$ be the (expanded) model as above and let $\mathbf{M}'$ be its restriction to the (interpretations of) all constants of all levels. Then evidently $\|C(d)\|^*_{\mathbf{M}} = \|C(d)\|^*_{\mathbf{M}'}$ for all formulas $C(d)$ processed in the construction of T (proof by induction on the complexity of C); in particular, $\|C_0(a_0)\|_{\mathbf{M}} = \|C_0(a_0)\|_{\mathbf{M}'}$ and obviously, $\mathbf{M}'$ is finite.

Corollary 1. *If there is a* $*$-*witnessed* $*$-*model* $\mathbf{M}$ *with* $\|C_0(a_0)\|^*_{\mathbf{M}} = v \in [0, 1]$ *then there is a finite* $*$-*model* $\mathbf{M}'$ *with* $\|C_0(a_0)\|^*_{\mathbf{M}'} = v$. *Consequently, if* $(\forall x)C_0(x)$ *is true in all finite models then it is true in all witnessed models* (*in particular, for Łukasiewicz logic* $(\forall x)C_0(x)$ *is then true in* all *models, see Section* 3 *for details*).

Example 2. We show that in Gödel fuzzy logic we can have a concept satisfiable in degree 1 in an infinite model but having value 0 in each finite model (and in each witnessed model). The example is $(\neg\forall R.A)\&(\neg\exists R.\neg A)$. Let N be the set of natural numbers, let $a = 0$, $\|R(m, n)\| = 1$ iff $m = 0$ and

$n \in N$, for $n \in N$ let $\|A(n)\| = \frac{1}{n+1}$. Then

$$\|(\forall x)(R(a,x) \to A(x))\| = \inf_n \|R(a,n) \to A(n)\| = 0, \quad \text{thus } \|\neg(\forall R.A)(a)\| = 1,$$

$$\|(\exists x)(R(a,x)\&\neg A(x)\| = \sup_n \|R(a,n)\&\neg A(n)\| = 0, \quad \text{thus } \|\neg(\exists R.\neg A)(a)\| = 1.$$

Now assume a model $\mathbf{M}$ in which an element a satisfies our concept in degree 1 and the formula $(\forall R.A)(a)$ is witnessed, i.e., for some $d \in M$, $\|R(a,d) \to A(d)\| = 0$. It follows $\|R(a,d)\| > 0$ and $\|A(d)\| = 0$ (Gödel implication!). Then $\|R(a,d) \wedge \neg A(d)\| = \|R(a,d) \wedge 1\| = \|R(a,d)\| > 0$, thus $\|(R(a,d) \wedge \neg A(d)\| = 0$ and $\|(\neg\exists R.\neg C)(a)\| = 0$, thus our concept has value 0, a contradiction.

Remark 3. Let T be a theory over $G\forall$ and φ a (closed) formula. Let $\|\varphi\|_T^G$ be $\inf\{\|\varphi\|_{\mathbf{M}}^G \mid \mathbf{M}$ a $[0,1]_G$-model of $T\}$. Observe that $\|\varphi\|_T^G$ is 0 or 1. Indeed, if there is an $\mathbf{M}$ such that $\|\varphi\|_M^G$ is an element of the open interval $(0,1)$ then applying an increasing 1–1 mapping of $[0,1]$ onto itself you can produce an isomorphic model $\mathbf{M}'$ of T in which $\|\varphi|_{\mathbf{M}'}^G$, is as small (positive) as you like; thus $\inf\{\|\varphi\|_{\mathbf{M}}^G|\mathbf{M}$ an $[0,1]_G^*$-model of $T\} = 0$. This applies to the infimum of degrees of satisfiability of a concept: it is 1 or 0.

3. Satisfiability, validity, subsumption of concepts

Definition 5. A concept C is $*$-*satisfiable* if there is an interpretation $\mathbf{M}$ such that $\|C(a)\|_{\mathbf{M}}^* = 1$ (a being a constant); C is $*$-*valid* if for all $\mathbf{M}$, $\|(\forall x)C(x)\|_{\mathbf{M}}^* = 1$; in particular, C is $*$-*subsumed* by D if $C \to D$ is $*$-valid.[5]

Note that C is $*$-valid iff for all $\mathbf{M}$, $\|C(a)\|_{\mathbf{M}}^* = 1$ where a is a constant (since $C(x)$ contains no constants). Here we allow no a priori axioms; if you would have some, a would have to be a new constant not occurring in your axioms. Our aim is to discuss the decidability of the three notions just introduced. How can we use the algorithm from the preceding section? What is the impact of *witnessing* quantified formulas?

First, clearly each *finite* model is $*$-witnessed, whatever your t-norm $*$ is (finite suprema are maxima, analogously infima). Also clearly, there are infinite models with some formulas not witnessed: if $\mathbf{M}$ interprets a unary predicate A, for all elements $u \in M$ the value $\|A(u)\|_{\mathbf{M}}$ is positive but the infimum over all u is 0 then in $\mathbf{M}$ the formula $(\forall x)A(x)$ is not witnessed. Similarly for $(\exists x)A(x)$, if $\sup \|A(u)\|_{\mathbf{M}}$ is bigger than each $\|A(u)\|_{\mathbf{M}}$.

Remark 4. Here we mention an *error* in [17]: In the proof of Lemma 13 point (1) part 4 [17, p. 10] the authors assume that in a model $(\mathfrak{A}, \tau)$ the formula $(\exists R.C)(x)$ has a value t and conclude that there is an element ξ in the model such that t is the minimum (conjunction) of the values of $R(x,\xi)$ and (ξ), i.e., that

[5] This is equivalent to the definition of [2] (Section 3.4). Our advantage is that we can express subsumption of C, D as validity of a formula of our language (thanks to properties of residuated implication) and speak on the degree of inclusion of C in D in an interpretation as the truth degree of the formula $(\forall x)(C(x) \to D(x))$. This appears to be possibly more natural than the approach of [17].

the formula $(\exists R.C)(x)$, thus $(\exists y)(R(x, y) \wedge C(y))$ has a witness. This may not be the case as we have seen. Below we shall see how to overcome this defect.

Lemma 3. *For the Łukasiewicz* t-*norm* $*$ *and each countable* $*$-*model* $\mathbf{M}$ (*of a given predicate language*) *there is a* $*$-*witnessed model* $\mathbf{M}'$ *such that* $\mathbf{M}$ *is a submodel of* $\mathbf{M}'$ *and* $\mathbf{M}$ *is elementarily equivalent to* $\mathbf{M}'$ (*in the sense that for any closed formula* α, $\|\alpha\|^*_{\mathbf{M}} = 1$ *iff* $\|\alpha\|^*_{\mathbf{M}'} = 1$).

Proof. Clearly it is enough to prove it for one fixed formula $(\exists x)\varphi(x)$ (since in Łukasiewicz logic, $(\forall x)\varphi(x)$ is equivalent to $\neg(\exists x)\neg\varphi(x)$) and then to iterate the construction countably many times. We use the extension of Ł∀ by rational truth constants $\bar{r}$ for any rational $r \in [0, 1]$ called RPL (rational Pavelka logic) in [6]. Assume that the language of $\mathbf{M}$ contains names of all elements of M and the crisply interpreted equality predicate ($r_=(a, b) = 1$ iff $a = b$). The *rational theory* $RTh(\mathbf{M})$ of $\mathbf{M}$ is the union over all closed formulas α of the sets

$$L_\alpha = \{\bar{r} \to \alpha \mid r \leqslant \|\alpha\|^{Ł}_{\mathbf{M}}, r \text{ rational}\},$$

$$U_\alpha = \{\alpha \to \bar{r}, \mid r \geqslant \|\alpha\|^{Ł}_{\mathbf{M}}, r \text{ rational}\}.$$

In particular, $RTh(\mathbf{M})$ contains these sets for each atomic closed formula $P(m_1, \ldots, m_n)$, P being a predicate and m_i being (names of) particular elements of M. $RTh(\mathbf{M})$ is a consistent theory over RPL∀ and adding a new constant c and the witnessing axiom $(\exists x)\varphi(x) \equiv \varphi(c)$ we get a conservative extension $\hat{T}$ of $RTh(\mathbf{M})$ (see Appendix B). By [6] 5.4.24 $\hat{T}$ has a standard model $\mathbf{M}'$. We may identify each element $m \in M$ with the corresponding element of $\mathbf{M}'$ (named with the same constant); this is a one–one relation due to the presence of crisp equality. And due to the presence of the axioms from L_α, U_α we see that $\|\alpha\|^{Ł}_{\mathbf{M}} = \|\alpha\|^{Ł}_{\mathbf{M}'}$ for each α. □

Remark 5. But e.g., for the Gödel t-norm this fails: think of a model $\mathbf{M}$ in which $\|(\forall x)P(x)\|^G_{\mathbf{M}} = 0$ but $\|P(u)\|^G_{\mathbf{M}} > 0$ for all $u \in M$. Then $\|(\exists x)\neg P(x)\|^G_{\mathbf{M}} = 0$ (Gödel negation), hence no $\mathbf{M}' \supseteq \mathbf{M}$ elementarily equivalent to $\mathbf{M}$ can contain a witness for $(\forall x)P(x)$. The same for product logic and any logic having Gödel negation.

Theorem 1. (1) *Over Łukasiewicz logic, a concept C is satisfiable iff it is satisfiable by a finite model iff the associated finite set* $prop(C(a)) \cup prop(T(C(a)))$ *is propositionally satisfiable.*

(2) *Over Łukasiewicz logic, a concept C is valid iff it is valid in all finite models iff the propositional theory* $prop(T(C(a)))$ *entails* $prop(C(a))$, *i.e. each evaluation e of propositional variables which is an Ł-model of* $prop(T(C(a)))$ *gives* $prop(C(a))$ *the Ł-value* 1.

Corollary 2. *Over Łukasiewicz logic, the* (*standard*) *satisfiability and* (*standard*) *validity of a concept are decidable problems*. (*See Appendix* B *for details*.)

Remark 6. (1) Propositional satisfiability of a formula (or a finite set of formulas) is NP in its size; but the size of the theory $prop(T)$ is *not* polynomial in the size of $C(a)$). Similarly for entailment and co-NP. Recall that satisfiability/validity of a concept over classical logic is PSPACE-complete; [16] generalizes

this for his fuzzy description logic (over the "minimalist" fuzzy logic). An analysis of the complexity of our algorithm is a matter of further investigation.

(2) Note that the "minimalist" logic of min, max, $1 - x$ is a sublogic of Łukasiewicz logic and the Kleene–Dienes implication is definable; thus our proof can be adapted to get proofs of (known) facts on the "minimalist" fuzzy ALC. But let us present here an example of the fact that the minimalist logic is sometimes counter-intuitive due to its weakness. Think of the following instance of a concept: all hotels near to the main square are expensive (clearly expressible as $(\forall R.C)(msquare)$. Imagine the ideal situation that for each hotel the degree of its being near to the main square equals to the degree of its being expensive (and that there is at least one hotel which is near to the main square in degree 0.5). Then the truth degree of $(\forall R.C)(msquare)$ should be 1 and it is so when you use residuated implication. But for Kleene–Dienes implication (i.e., for the definition of the truth degree of our instance following [15]) it is 0.5, which appears to be strange.

(3) Let us mention in passing the paper [3] showing nicely how two definitions equivalent in classical logic may differ in fuzzy logic: classically, a formula $C(a)$ is satisfiable iff it there is a model in which it has the value 1, equivalently if there is a model in which it has a *positive* value (since there are just two values, 0 and 1). For (the minimalist) fuzzy logic this gives two different notions, first being PSPACE-complete but the second being decidable in linear time.

(4) Clearly there are concepts satisfiable over Łukasiewicz but not our classical logic; the simplest example is the concept $A \equiv \neg A$ (i.e., $(A \to \neg A)\&(\neg A \to A)$).

4. On arbitrary continuous t-norms

In the preceding section we focused our attention on Łukasiewicz logic; here we formulate consequences of our observations for a logic given by an arbitrary fixed continuous t-norm $*$. Here the reader is assumed to know the (Mostert–Shields) representation of $*$ as an ordered sum of copies of Łukasiewicz, Gödel and product t-norms, as mentioned in Appendix A. The ordered set of copies may not have a first element; if it does have a first element and if it is Łukasiewicz we say that $*$ *begins* by Ł. Every $*$ not beginning by Ł has Gödel negation ($\neg 0 = 1$, $\neg x = 0$ for $x > 0$). The following is a very useful observation (see [8]):

Lemma 4. (1) *If $*$ begins by Ł then, for any propositional formula φ, φ is $*$-satisfiable iff it is Ł-satisfiable.*

(2) *If $*$ does not begin by Ł then for each φ, φ is $*$-satisfiable if it is satisfiable in the classical Boolean logic.*

Theorem 2. *Let $*$ begin by Ł. Then the following are equivalent, for any concept C:*

(1) *C is satisfiable by a witnessed $*$-model,*
(2) *C is satisfiable by a finite $*$-model,*
(3) *$prop(C(a)) \cup prop(T(C(a)))$ is $*$-satisfiable*
(4) *$prop(C(a)) \cup prop(T(C(a)))$ is Ł-satisfiable*

Theorem 3. *If $*$ does not begin by Ł then the following are equivalent for any concept C:*

(1)–(3) *as in the preceding theorem*

(4) *$prop(C(a)) \cup prop(T(C(a)))$ is boolean satisfiable.*

Theorem 4. *For any continuous* t-*norm* $*$ *and any concept C, the following are equivalent*:

(1) *C is valid in all witnessed* $*$-*models*,

(2) *C is valid in all finite* $*$-*models*,

(3) *prop*$(T(C(a)))$ $*$-*entails prop*$(C(a))$.

All these theorems are immediate consequences of our lemmas.

Corollary 3. *For an arbitrary continuous* t-*norm* $*$, *witnessed* $*$-*satisfiability and witnessed* $*$-*validity of a concept is decidable*.

This is because propositional $*$-satisfiability and propositional $*$-entailment are decidable, see Appendix B.

Remark 7. Apparently finite models are more important for applications of description logic than infinite models; thus our result on decidability of satisfaction and validity of concepts in finite models is of main importance. On the other hand, for logical investigations infinite models are of high importance and our results show that one has to be careful: in Example 2 we have seen a concept which is not satisfiable in any finite model (thus not satisfiable in any witnessed infinite model) but is satisfiable in an infinite non-witnessed model. Witnessed models appear not to have been investigated in the literature and we have seen that they are indeed important. Their properties will be further studied in the paper [7]. Our investigation leads to several open problems and research tasks; we list some of them.

Problems. (1) Analyze computational complexity results from existing papers on (fuzzy) description logic and their impact to our approach. Are the "classical" problems of description logic in our setting also PSPACE-complete?

(2) Allow *axioms* (terminological axioms and assertions) similarly as in the classical description logic and in the minimalist fuzzy description logic. Analyze the problem of *degree of satisfaction* of a concept, i.e., $\inf_{\mathbf{M}}\{\|C(a)\|^{Ł}_{\mathbf{M}}\}$ and dually the problem of *degree of validity*.

(3) Some proofs of decidability of problems of description logic used the fact that they can be expressed in classical predicate logic having only two object variables, which is decidable. Is Łukasiewicz predicate logic (Gödel, product logic, BL) with only two object variables decidable? (Observe that Łukasiewicz with two variables does *not* have finite model property: consider the formula

$$(\exists x)(P(x) \equiv \neg P(x))\&(\forall x)(\exists y)(P(x) \equiv (P(y)\&P(y))).$$

5. Conclusion

Admittedly, the minimalist fuzzy logic is rather simple and shares some nice properties of classical logic, as definability of $\forall$ from $\exists$ and vice versa and the possibility of expressing every formula equivalently by a formula where negation occurs only with atoms. This is not the case in general for the continuous t-norm

based logic (BL); but BL offers much richer expressive possibilities and finer distinctions. We showed above some weaknesses of the description logic based on the minimalist fuzzy means. All this justifies the investigation of description logics based on richer fuzzy logics. We hope to have shown two things: description logic can profit from advanced fuzzy logic getting richer expressive possibilities (but still decidable!) and fuzzy logic may profit from description logic by getting new problems and inspirations.

Appendix A.

This is a very short survey of the basic fuzzy logic BL (as developed in my monograph [6]) and its standard semantics given by continuous t-norms. The following are two starting assumptions made:

(1) The real unit interval $[0, 1]$ is taken to be the *standard set of truth values*; *comparative notion of truth*.
(2) The logic is *truth-functional*, i.e., connectives are interpreted via their truth functions.
(3) *Continuous* t-*norms* are taken as possible truth functions of *conjunction*.

A t-*norm* is a binary operation $*$ on $[0, 1]$ (i.e. $* : [0, 1]^2 \to [0, 1]$) satisfying the following conditions:

(i) $*$ is commutative and associative, i.e., for all $x, y, z \in [0, 1]$,

$$x * y = y * x, \quad (x * y) * z = x * (y * z),$$

(ii) $*$ is non-decreasing in both arguments, i.e.

$x_1 \leqslant x_2$ implies $x_1 * y \leqslant x_2 * y$,

$y_1 \leqslant y_2$ implies $x * y_1 \leqslant x * y_2$,

(iii) $1 * x = x$ and $0 * x = 0$ for all $x \in [0, 1]$.

The following are our most important examples of continuous t-norms:

(i) *Łukasiewicz* t-*norm*: $x * y = \sup(0, x + y - 1)$,
(ii) *Gödel* t-*norm*: $x * y = \min(x, y)$,
(iii) *Product* t-*norm*: $x * y = x.y$ (product of reals).

Let $*$ be a continuous t-norm. Then there is a unique operation $x \Rightarrow y$ satisfying, for all $x, y, z \in [0, 1]$, the condition $(x * z) \leqslant y$ iff $z \leqslant (x \Rightarrow y)$, namely $x \Rightarrow y = \max\{z \mid x * z \leqslant y\}$.

The operation $x \Rightarrow y$ is called the *residuum* of the t-norm.

The following operations are residua of the three t-norms above: $x \Rightarrow y = 1$ for $x \leqslant y$; for $x > y$:

(i) *Łukasiewicz implication*: $x \Rightarrow y = 1 - x + y$
(ii) *Gödel implication*: $x \Rightarrow y = y$
(iii) *Goguen implication*: $x \Rightarrow y = y/x$ (residuum of product conjunction).

The *Representation theorem* (by Mostert and Shields): says that each continuous t-norm is an ordered sum of copies of Łukasiewicz, Gödel and product t-norms. (See e.g. [6] for details.)

The basic many-valued logic is a symbolic logical system. Formulas are built from propositional variables, connectives & (strong conjunction) and $\to$ (implication) and from the truth constant $\bar{0}$ (falsity).

Further connectives are defined as follows:

$$\begin{array}{rcl} \varphi \wedge \psi & \text{is} & \varphi \& (\varphi \Rightarrow \psi), \\ \varphi \vee \psi & \text{is} & ((\varphi \to \psi) \to \psi) \wedge ((\psi \to \varphi) \to \varphi), \\ \neg\varphi & \text{is} & \varphi \to \bar{0}, \\ \varphi \equiv \psi & \text{is} & (\varphi \to \psi) \& (\psi \to \varphi). \end{array}$$

An *evaluation of propositional variables* is a mapping e assigning to each propositional variable p its truth value $e(p) \in [0, 1]$.

This extends, given a continuous t-norm $*$ and its residuum $\Rightarrow$:

$$e(\bar{0}) = 0,$$

$$e(\varphi \to \psi) = (e(\varphi) \Rightarrow e(\psi)),$$

$$e(\varphi \& \psi) = (e(\varphi) * e(\psi)).$$

Note that the truth function of $\wedge$ is minimum and that of $\vee$ is maximum, independently from the choice of $*$. (But of course, for a general φ, $e(\varphi)$ is given by the evaluation e and by the choice of the continuous t-norm $*$.) A formula φ is a $*$-tautology if $e(\varphi) = 1$ for each e. Furthermore, e is a $*$-model of a set S of formulas if $e(\varphi) = 1$ for all elements φ of S.

For more detailed information see my monograph [6]; you will find there axioms of the basic fuzzy predicate logic, general semantics based on BL-algebras (general algebras of truth-functions) and completeness theorems (see also [5]). These things will not be necessary in the present paper.

Now turn to the basic fuzzy predicate calculus.

The basic many-valued predicate logic works with a *predicate language* consisting of *predicates* $P, Q, \ldots$, each together with its *arity* and *object constants* $c, d, \ldots$. *Logical symbols are object variables* $x, y, \ldots$, *connectives* $\&, \to$, *truth constants* $\bar{0}, \bar{1}$ and *quantifiers* $\forall, \exists$. Other connectives ($\wedge, \vee, \neg, \equiv$) are defined as in prop. calculus. *Terms* are object variables and object constants.

Atomic formulas have the form $P(t_1, \ldots, t_n)$ where P is a predicate of arity n and $t_1, \ldots, t_n$ are terms. If φ, ψ are formulas and x is an object variable then $\varphi \to \psi$, $\varphi \& \psi$, $(\forall x)\psi$, $(\exists x)\varphi$, $\bar{0}$, $\bar{1}$ are formulas; each formula results from atomic formulas by iterated use of this rule.

A *fuzzy structure* for our language has the form $\mathbf{M} = \langle M, (r_P)_P, (m_c)_c \rangle$ where $M \neq \emptyset$, for each n-ary predicate P, r_P is a fuzzy n-ary relation $r_P : M^n \to [0, 1]$ on M and for each object constant c, m_c is an element of M.

Let $*$ be a continuous t-norm and $\Rightarrow$ its residuum. An $\mathbf{M}$-*evaluation* v of object variables assigns to each object variable x an element $v(x)$ of M denoted also $\|x\|_{M,v}$; for a constant c let $\|c\|_{M,v} = m_c$. (The value of a variable or a constant given by $\mathbf{M}$ and v.)

The truth value $\|\varphi\|^*_{M,v}$ of a formula φ given by the interpretation $\mathbf{M}$, the t-norm $*$ and the evaluation v of variables is defined inductively as follows:

$$\|P(t_1, \ldots, t_n)\|^*_{M,v} = r_P(\|t_1\|_{M,v}, \ldots, \|t_n\|_{M,v}),$$

$$\|\varphi \to \psi\|^*_{M,v} = \|\varphi\|^*_{M,v} \Rightarrow \|\psi\|^*_{M,v},$$

$$\|\varphi \& \psi\|^*_{M,v} = \|\varphi\|^*_{M,v} * \|\psi\|^*_{M,v},$$

$\|\bar{0}^*\|_{M,v}$ is 0,

$\|(\forall x)\varphi\|^*_{M,v} = \inf\{\|\varphi\|^*_{M,v'} | v \equiv_x v'\}$,

$\|(\exists x)\varphi\|^*_{M,v} = \sup\{\|\varphi\|^*_{M,v'} | v \equiv_x v'\}$

where $v \equiv_x v'$ means that for all variables y different from x the value $v(y)$ equals to the value y.

In words, the value of an atomic formula is the degree in which the tuple of values of its terms is in the fuzzy relation interpreting its predicate. Connectives are interpreted using the truth functions. The value of an universally quantified formula is the infimum of its instances and similarly for an existentially quantified formula and supremum. Finally, writing $\|\varphi\|^*_M$ we mean the infimum of $\|\varphi\|^*_{M,v}$ over all evaluations v.

A formula φ of our language is a $*$-*tautology* if $\|\varphi\|^*_M = 1$ for each fuzzy structure **M**.

Caution. In classical logic, $(\forall x)\varphi$ is equivalent to $\neg(\exists x)\neg\varphi$ and similarly the existential quantifier is definable from the universal quantifier. This fails in fuzzy logic, except for the Łukasiewicz t-norm. (Cf. Example 2 in Section 2.)

For more detailed information I again refer to [6]—general semantics (interpretations over BL-algebras), completeness theorem as well as some non-axiomatizability theorems and other things, not necessary for the present paper, except explicitly stated.

Appendix B.

First we discuss computational complexity of propositional $*$-satisfiability and $*$-entailment. Then we prove a theorem on witnessing axioms used for a proof of Lemma 3 above.

The fact that the set of all $*$-satisfiable formulas is NP-complete was first proved for Łukasiewicz $*$ by Mundici [13] (see also [6] for an alternative proof). Thus for each $*$, we get the same result by Lemma 4 above. The co-NP completeness of all $*$-tautologies was proved for Łukasiewicz in [13]; for Gödel and product in [6] and for arbitrary $*$ in [9] as well as in [8]. For $*$-entailment the result seems to be new (but easy from known facts).

Theorem B.1. *For each continuous* t-*norm* $*$, *the set of all pairs* (φ, ψ) *of* (*propositional*) *formulas such that* φ $*$-*entails* ψ *is co-NP-complete*.

Proof. For Łukasiewicz inspect in [6] the proof of the fact that Ł-tautologies are co-NP: modify the algorithm assigning to each formula ψ a mixed integer problem such that solutions of the problem give evaluation e such that $e(\psi) < 1$, getting a very similar algorithm assigning to each pair φ, ψ an MIP-problem whose solutions give $e(\varphi) = 1$ and $e(\psi) < 1$. For Gödel use classical deduction theorem (valid for $G\forall$) reducing entailment of ψ from φ to tautologicity of $\varphi \to \psi$. For product inspect the proof in [6] analogously to Łukasiewicz above. Having all this inspect the proof of Haniková from [9,8] and make corresponding modifications. □

In the rest the reader is assumed to know basic facts on theories over the fuzzy predicate logic BL$\forall$ and stronger logics. Recall that T' is a *conservative extension* of T if $T' \supseteq T$ and each formula φ in the language of T provable in T' is provable in T.

Lemma B.1. *Each theory T over Łukasiewicz logic* Ł∀ *has a conservative extension T containing witnessing axioms for all closed formulas beginning by a quantifier.*

Proof. Clearly, it is enough to show that adding just one witnessing axiom is conservative (and then iterate); moreover, it suffices to discuss only ∃ (due to definability of ∀ from ∃ in Ł∀). Thus let c be a new constant and assume

$$T, (\exists x)\varphi(x) \to \varphi(c) \vdash \alpha,$$

where α does not contain c. Then for some n,

$$T \vdash [(\exists x)\varphi(x) \to \varphi(c)]^n \to \alpha,$$

thus for a new variable y,

$$T \vdash (\forall y)([(\exists x)\varphi(x) \to \varphi(y)]^n \to \alpha),$$

(just replace c by y in the proof)

$$T \vdash (\exists y)[(\exists x)\varphi(x) \to \varphi(y))^n] \to \alpha; \text{ then by [6] 5.1.18(10)},$$

$$T \vdash [(\exists y)((\exists x)\varphi(x) \to \varphi(y)]^n \to \alpha \text{ and, by [6], 5.4.15(i)},$$

$$T \vdash [(\exists x)\varphi(x) \to (\exists y)\varphi(y)]^n \to \alpha \text{ (here Ł}\forall\text{-provability is used!), and}$$

$$T \vdash \alpha, \text{ since the formula [...] is evidently provable.} \qquad \square$$

Our Example 2 above shows that adding a witnessing axiom may be non-conservative over $G\forall$ (and over $\Pi\forall$-the same example). Thus an analogon of Lemma 3 does not hold for $G\forall$, $\Pi\forall$; one has to work with witnessed models. (Recall once more that each finite model is witnessed; also if you work with Gödel logic over a fixed finite subset of truth values ($v \subseteq [0, 1], 0 \in v, 1 \in v$) all models (finite or infinite) are witnessed. Also note in passing that BL∀ (and hence each stronger logic) has the following conservation property.

Lemma B.2. *Let T be a theory such that* $T \vdash (\exists x)\varphi(x)$, *let c be a new constant. Then* $T \cup \{\varphi(c)\}$ *is a conservative extension of T.*

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